COVID-19 Tweet Sentiment Classification

A Machine Learning Approach Using TF-IDF and Linear Classifiers

Springboard Data Science Career Track | Capstone Three Final Report

1. Problem Statement

The rapid spread of COVID-19 generated an unprecedented volume of social media discourse. Understanding how the public emotionally responded to the pandemic through tweet sentiment offers valuable insight for public health communicators, policymakers, and researchers. This project addresses the following question:

> ***Can machine learning models accurately classify the sentiment of COVID-19-related tweets into Positive, Neutral, or Negative categories using only the tweet text?***

Accurate sentiment classification enables organizations to monitor public mood in real time, identify misinformation-driven panic, and tailor their messaging accordingly.

2. Data & Methodology

2.1 Dataset

The dataset used is the Corona NLP dataset, a publicly available collection of tweets captured during the COVID-19 pandemic. The test split contains 3,798 labeled tweets across five original sentiment categories that were consolidated to three.

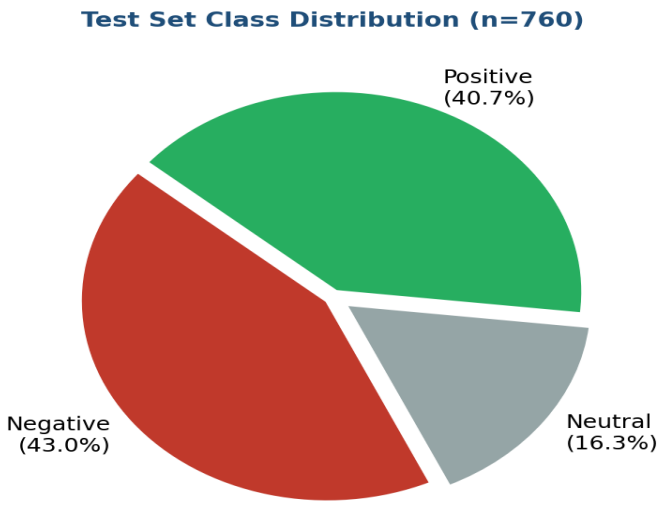

Figure 1: Test set class distribution after sentiment consolidation (n=760). Negative and Positive classes are nearly balanced; Neutral is the minority class.

2.2 Data Cleaning & Preprocessing

The raw tweet text underwent several cleaning steps to reduce noise and standardize inputs:

- Sentiment consolidation: Five original labels mapped to three classes (Positive, Neutral, Negative).
- Text normalization: All text was lowercased.
- URL removal: HTTP/HTTPS links were stripped as they carry no semantic sentiment value.
- Mention and hashtag removal: Twitter-specific tokens (@mentions, #hashtags) were removed.
- Special character filtering: Non-alphabetic characters were removed, retaining only letters and spaces.
- Whitespace normalization: Consecutive spaces collapsed to single spaces.

Two tweets with missing text values were identified and handled with empty-string imputation to prevent data loss.

2.3 Feature Engineering

TF-IDF (Term Frequency-Inverse Document Frequency) was applied to convert cleaned tweet text into numerical features. Key configuration: 5,000 max features, unigrams and bigrams (1,2) n-gram range, English stop words removed. This produced a sparse matrix of 3,038 training and 760 test samples, each represented as a 5,000-dimensional vector.

2.4 Train / Test Split

An 80/20 stratified split (random_state=42) was applied, preserving class distribution in both subsets.

3. Modeling

3.1 Models Evaluated

Three classification algorithms were evaluated and tuned with 5-fold GridSearchCV, each representing a different modeling philosophy for text classification:

- Logistic Regression: A probabilistic linear classifier well-suited to high-dimensional sparse data.
- Linear SVC: A margin-maximizing linear classifier. Hyperparameter search focused on regularization C.
- Multinomial Naive Bayes: A generative probabilistic model based on word frequency.

3.2 Model Comparison

Model	Accuracy	Macro F1	Best Params
Logistic Regression ✓	67.8%	0.62	C=10, liblinear
LinearSVC	67.1%	0.62	C=1
Multinomial Naive Bayes	59.9%	0.44	alpha=1.0

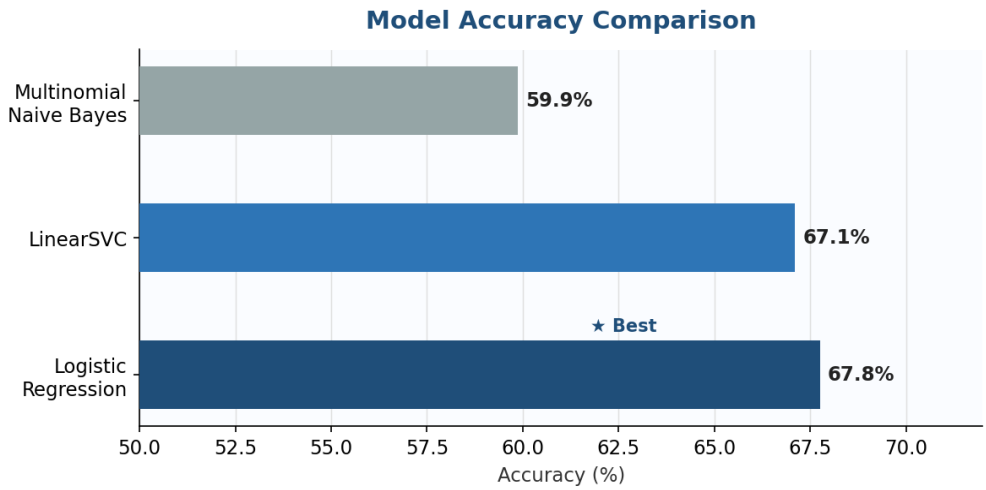

Figure 2: Accuracy comparison across all three models on the held-out test set. Logistic Regression achieves the highest accuracy at 67.8%.

Logistic Regression emerged as the best-performing model (67.8%), narrowly outperforming LinearSVC (67.1%). Multinomial Naive Bayes trailed significantly at 59.9%.

4. Key Findings

4.1 Classification Performance by Class

Class	Precision	Recall	F1-Score	Support

Negative	0.70	0.78	0.74	327
Neutral	0.57	0.35	0.43	124
Positive	0.68	0.71	0.69	309
Weighted Avg	0.67	0.68	0.67	760

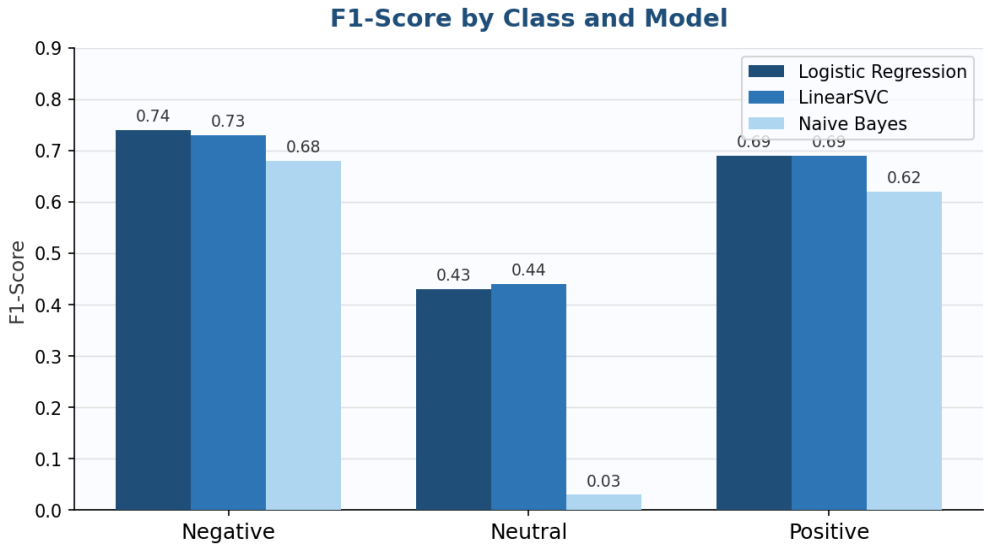

Figure 3: F1-scores per sentiment class for all three models. Logistic Regression leads on Negative and ties on Positive. The Neutral class underperforms across all models.

The model demonstrates notably stronger performance on clearly polarized classes. Negative tweets achieved the highest F1-score (0.74) with strong recall (0.78). The Neutral class was most challenging (F1=0.43), as neutral language frequently overlaps with both positive and negative vocabulary.

4.2 Confusion Matrix Analysis

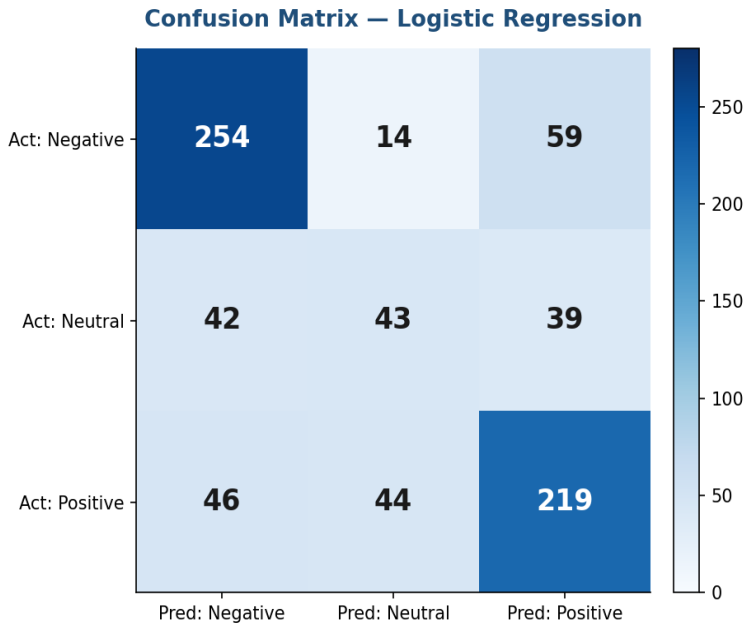

Figure 4: Confusion matrix for the best model (Logistic Regression, C=10). The diagonal represents correct predictions. The Neutral row shows the most misclassifications.

Analysis of the confusion matrix revealed:

- 254 of 327 Negative tweets correctly classified (78% recall)
- 219 of 309 Positive tweets correctly classified (71% recall)
- Only 43 of 124 Neutral tweets correctly classified (35% recall)
- Most common misclassification: Neutral tweets predicted as Positive or Negative

4.3 Why Linear Models Outperformed Naive Bayes

TF-IDF produces a high-dimensional sparse feature space that is well-suited for linear classifiers. Logistic Regression and LinearSVC leverage geometric margin separation, while Multinomial Naive Bayes relies on a feature independence assumption that breaks down in natural language. This explains the ~8 percentage point accuracy gap.

5. Recommendations

Recommendation 1: Deploy the Logistic Regression Model for Real-Time Monitoring

The tuned Logistic Regression classifier (C=10, liblinear, TF-IDF 5,000 features, bigrams) should be deployed as a baseline sentiment monitoring tool for COVID-19 social media feeds. At 67.8% accuracy with strong recall on the Negative class (78%), it provides a practical signal for detecting surges in negative public sentiment. The model is lightweight and fast, making it suitable for streaming inference on high-volume tweet feeds.

Recommendation 2: Treat Neutral Classification as a Separate Research Problem

The Neutral class consistently underperforms across all three models (F1 range: 0.03–0.44). Organizations should apply a binary interpretation — flag tweets as Positive or Negative — and treat low-confidence predictions as Neutral. Consider a two-stage model: a binary classifier (Positive vs. Not-Positive) followed by a second stage (Negative vs. Neutral).

Recommendation 3: Invest in Transformer-Based Models for Production

The ~32% error rate reflects fundamental limits of bag-of-words features. Large language models such as BERTweet (BERT pretrained on Twitter data) are expected to significantly outperform TF-IDF + linear classifiers because they encode contextual and semantic meaning, not just word frequency. Fine-tuning BERTweet on this dataset is the highest-ROI next step.

6. Ideas for Further Research

- Transformer fine-tuning: Fine-tune BERTweet or COVID-Twitter-BERT on the full Corona NLP training split.
- Temporal analysis: Explore how sentiment shifted over time as the pandemic evolved using tweet date metadata.
- Geographic sentiment mapping: Leverage the Location field to map sentiment by region.
- Class imbalance handling: Experiment with SMOTE oversampling or class_weight='balanced' to improve Neutral recall.
- Ensemble methods: Combine Logistic Regression and LinearSVC predictions via soft voting.
- Expanded hyperparameter search: Wider C ranges, different max_features, and sublinear TF scaling.

7. Conclusion

This project demonstrated that TF-IDF combined with Logistic Regression provides a strong, efficient baseline for COVID-19 tweet sentiment classification, achieving 67.8% accuracy on a three-class problem. The model reliably identifies negative and positive sentiment — making it actionable for real-time public health monitoring — while its primary limitation lies in classifying neutral content.

The clearest path to improvement is adopting transformer-based language models fine-tuned on Twitter-specific data. The current model, however, is immediately deployable and provides meaningful signal for public health communication strategy.